NetLift: A Causal Decision System for Incremental Profit Optimization

Phase 1: Project Planning and Management

Digital Egypt Pioneers Initiative (DEPI), Microsoft Machine Learning Engineer Track

October 2026

Team leader: Mahmoud Nadeem

Team member	Role
Mahmoud Nadeem	Team Leader; Advanced Models and Experiment Tracking
Aliaa Gomaa	Data Engineering and Experiment Validity
Mohammed Sherif	Uplift Modelling Core and Dashboard
Ali Alaa	Evaluation, Statistics and Decision Layer
Mazen Mahmoud	MLOps, Backend and Deployment

Repository: https://github.com/mahmoud-nadeem/netlift

Contents

1. Project Proposal
2. Project Plan
3. Task Assignment, Roles and Working Agreement
4. Risk Assessment and Mitigation Plan
5. Key Performance Indicators (KPIs)

1. Project Proposal

The approved project proposal: overview, objectives and scope. Its original week-by-week plan and risk table are superseded by sections 2 and 4, which follow the fixed DEPI deadlines.

Objective

To build and deploy a machine-learning system that predicts, for each customer, how much a marketing or retention intervention changes that customer's behaviour, and to use that estimate to allocate a fixed campaign budget so that it produces the largest possible number of additional conversions.

The project is deliberately built around a question that a standard classification model cannot answer. A churn or response model predicts who is likely to convert; it cannot tell whether the contact caused the conversion. NetLift estimates the causal effect of the contact itself, which is the quantity a business actually spends money against.

Measurable objectives. The project KPIs are defined in section 5.

K1 is the project. Everything else supports it. If K1's interval excludes zero, the central claim holds. If it contains zero, the report says so and the project becomes an investigation into why, which is why K1 is scheduled at Milestone 4 and not at the end.

Description

Companies allocate retention and promotion budgets using models that rank customers by predicted conversion or predicted churn. Those models are frequently accurate and the campaigns built on them frequently lose money, because the ranking contains no information about causation. Customers at the top of a “likely to convert” list often include the customers who would have converted with no offer at all, so the discount they receive is margin given away for nothing.

Uplift modelling addresses this directly by estimating the Conditional Average Treatment Effect: the difference between what happens if a customer is contacted and what happens if the same customer is not. The population divides into four groups, and two of them cost money when targeted:

Segment	If contacted	If not contacted	Targeting implication
Persuadables	Converts	Does not convert	The only group worth spending budget on
Sure Things	Converts	Converts	Zero effect: the offer is wasted margin
Lost Causes	Does not convert	Does not convert	Zero effect: wasted contact cost
Sleeping Dogs	Does not convert	Converts	Negative effect: contacting them destroys value

A response model cannot separate Persuadables from Sure Things, because it only ever observes conversions among contacted customers. NetLift makes that separation, and additionally identifies the Sleeping Dogs so that they can be suppressed from the campaign entirely, a behaviour no conventional targeting engine can express.

Because the individual treatment effect is never observable for any single customer (only one of the two outcomes ever happens), the project cannot be validated with accuracy, F1 or ROC-AUC. It is validated instead with uplift-specific metrics computed against a randomized control group, each reported with bootstrap confidence intervals, and confirmed by a negative-control experiment. This constraint is the intellectual core of the project and shapes its data strategy, its evaluation design and its testing plan.

Data. Two public datasets from genuine randomized experiments are used, which is what makes the causal claim verifiable rather than assumed:

- **Criteo-UPLIFT ,** 13,979,592 rows, 12 anonymised numeric features, treatment ratio 0.85, average visit rate 0.046992 and average conversion rate 0.00292, with visit and conversion as binary targets. Released with Diemert, Betlei, Renaudin and Amini, “A Large Scale Benchmark for Uplift Modeling”, AdKDD Workshop at KDD 2018. Used for the primary results.
- **Hillstrom MineThatData ,** 64,000 customers randomly split into equal thirds across a Mens e-mail campaign, a Womens e-mail campaign and a no-e-mail control, with visit, conversion and spend recorded over the following two weeks. Used for pipeline development, for multi-treatment modelling and for revenue (continuous-outcome) uplift.

Deliverable. A validated data pipeline, a set of uplift models benchmarked against the response model a business would otherwise use, an evaluation harness with confidence intervals, a budget-constrained decision layer that includes an explicit “insufficient evidence” outcome, a campaign policy simulator that reports incremental profit, and a deployed scoring service with a dashboard and drift monitoring.

A system, not a model, and stated honestly. Criteo and Hillstrom are fixed historical datasets: there is no live audience to score, no target list that will be handed to a campaign system, and no outcome window that will close while we wait. Every outcome was recorded years ago and is already in the file. What this project builds is therefore a simulated campaign replay: a slice of the held-out test split stands in for next month’s audience, the system defines a scenario with its cost and margin, scores that slice without reading its outcomes, allocates the budget, holds out a randomized share of the selected targets, and only then reveals the recorded outcomes and measures what the decision actually produced, against the response-model, blanket and random policies at an identical budget.

This measures the decision rather than the ranking, which a Qini curve alone cannot do, and it is possible precisely because the data is historical: the counterfactual is already recorded, so nothing has to be waited for. It is also what makes five distinct roles necessary rather than nominal, it adds scenario and assignment records, holdout logic, the outcome-reveal boundary and the test that enforces it, and the measurement job.

What a live deployment would additionally require, a permanently maintained randomized holdout, outcome-window measurement, and the delayed-feedback handling that goes with them, is specified separately in the main document and is explicitly not claimed as work for this semester.

What the project estimates. The intention-to-treat effect, the effect of being selected for the campaign, because that is what the experimental design identifies and what a targeting decision actually controls. The effect on customers who genuinely received and registered the intervention would be larger, and estimating it would require exposure data that neither dataset contains. This is stated explicitly rather than left implicit.

Tools & Design

System design. A layered architecture with a strict separation between the offline training pipeline and the online scoring service:

- versioned ingestion, schema validation, duplicate removal, a randomization-integrity report, an automated leakage audit, and a train/validation/test split stratified jointly on treatment and outcome.
- three mandatory baselines (random targeting, blanket targeting, response model), then meta-learner uplift estimators, then advanced challengers, every run tracked in MLflow.
- Qini and uplift curves, AUUC, uplift@k and cumulative gain, all with bootstrap confidence intervals, plus a permuted-treatment negative control and a suite of behavioural invariance tests.
- converts an uplift score into an action under a cost structure, suppresses customers with confidently negative uplift, and returns an explicit abstention verdict when the evidence is too thin to support a recommendation.
- a FastAPI service loading the champion model from the MLflow registry by version, containerised with Docker and deployed on Azure, with a Streamlit dashboard and drift monitors on top.

The service is API-first: scoring and policy simulation are exposed as documented endpoints independent of the dashboard, so any other application could consume them without changes to the ML side.

Design artefacts to be produced. Use case diagram; functional and non-functional requirements; software architecture diagram; ER diagram with logical and physical schema (campaign, treatment assignment, holdout membership, scoring run, score, model version, outcome observation); context and level-1 data flow diagrams; sequence diagrams for batch scoring, policy simulation and retraining; activity diagram for campaign planning; state diagrams for the campaign and model lifecycles; class diagram of the pipeline object model; wireframes and UI guidelines; component and deployment diagrams; API documentation; test plan and test cases.

Technologies to Use

Area	Technologies
Language & data	Python 3.11, pandas, NumPy, PyArrow / Parquet
Machine learning	scikit-learn, LightGBM, XGBoost
Uplift / causal inference	scikit-uplift, CausalML (meta-learners, uplift forests)
Deep learning	PyTorch: shared-representation treatment-effect network
Statistics & explainability	SciPy, statsmodels, SHAP, bootstrap resampling

Area	Technologies
Hyperparameter search	Optuna, optimised against a Qini objective
Experiment tracking	MLflow tracking and model registry
Visualization	Matplotlib, Plotly
API & serving	FastAPI, Uvicorn, Pydantic
Frontend	Streamlit dashboard (campaign planner, policy comparison, model evidence, monitoring)
Storage	PostgreSQL, Azure Blob Storage
Cloud	Azure Machine Learning endpoint, Azure Container Registry, Application Insights
Containers & CI/CD	Docker, GitHub Actions
Testing	pytest: unit, data-validation, model-behaviour, integration and API tests
Documentation	Markdown, MkDocs or Sphinx, auto-generated OpenAPI

2. Project Plan

This is the committed project plan required by the DEPI documentation checklist: timeline (Gantt chart), milestones, deliverables and resource allocation. It is the version of record. Where it disagrees with any slide, message or older document, this file wins. How the team works day to day is in `working-agreement.md`.

Revised 13 October 2026. The first version planned in programme weeks, with seven milestones and a final presentation in Week 21. DEPI has since published fixed deadlines: six phases, final submission on **4 December 2026**. This revision plans against those dates.

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1. Deadlines

DEPI's deadlines are fixed and external. Every GitHub milestone carries its DEPI date as the due date. Internally, everything for a deadline is in an open pull request by **23:59 the day before**, so the deadline day itself is for review, merging and uploading only.

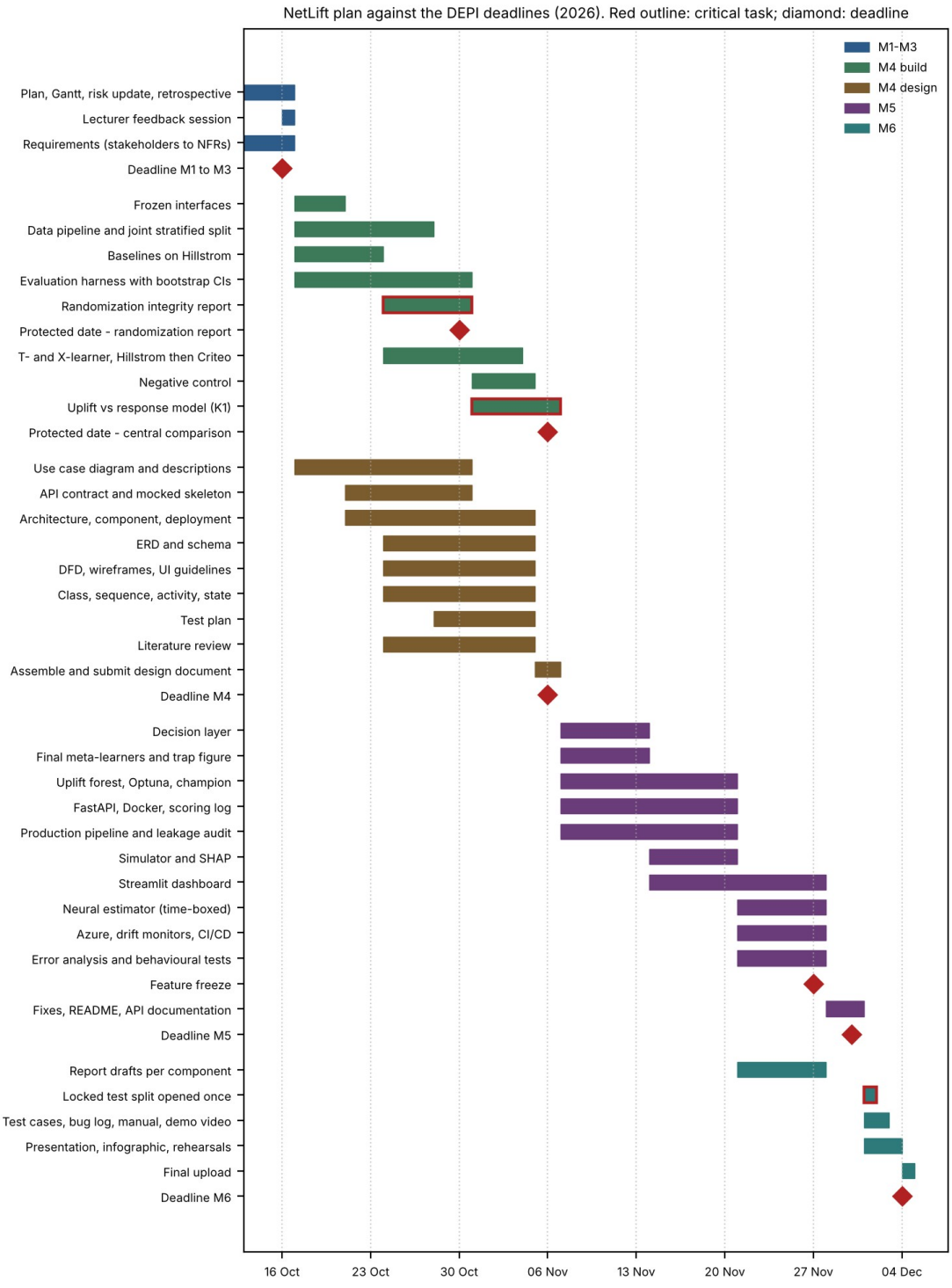
GitHub milestone	DEPI phase	DEPI deadline	Internal deadline
M1 - Project Planning and Management	Project Planning & Management	Fri 16 Oct 2026	Thu 15 Oct
M2 - Literature Review	Literature Review	Fri 16 Oct 2026	Thu 15 Oct
M3 - Requirements Gathering	Requirements Gathering	Fri 16 Oct 2026	Thu 15 Oct
M4 - System Analysis and Design	System Analysis & Design	Fri 6 Nov 2026	Thu 5 Nov
M5 - Implementation	Implementation (Source Code & Execution)	Mon 30 Nov 2026	Fri 27 Nov (feature freeze)
M6 - Testing, Final Presentation and Reports	Final Presentation & Testing & Reports	Fri 4 Dec 2026	Thu 3 Dec

The instructor confirmed on 13 October that the Literature Review phase is a feedback session with him on 16 October, not a graded deliverable. The written literature review therefore moves to M4, where it informs the design and later becomes the related-work section of the final report.

DEPI requires all documents on GitHub. The complete repository is pushed to the company repository at every deadline, not only at the end.

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2. Timeline



Project timeline (Gantt chart)

Two protected dates

Friday 30 October, randomization integrity report on both datasets. Standardized mean differences between arms and a treatment-predictability AUC. If the randomization is not what the publications claim, every later result rests on it, and the team must know before modelling, not after.

Friday 6 November, uplift versus response model at equal budget, with an interval (KPI K1). A response model is the honest baseline. If uplift modelling does not beat it on this data, that is a finding the project

reports and investigates. Found on 6 November there are three weeks to investigate it; found on 1 December there is nothing to say about it.

Everything else in the timeline can slip by a few days without damage. These two cannot.

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3. What changed, and what was cut

The final submission moved from mid-January to 4 December: roughly six weeks fewer. Three things change as a result.

Testing is no longer a phase of its own. Tests are written with the code they test, during M5, and every pull request keeps CI green. The M6 window is for opening the locked test split once, writing the test report, and the final presentation.

Implementation and design overlap. M4 is a design milestone, but the data pipeline, the split, the evaluation harness and the first estimators are built inside it, because the central comparison is due at its end.

Scope. Decided against the specification's scope-control order:

Item	Decision
Every MUST HAVE in the specification	Kept, unchanged
Confidence intervals, negative control, locked test split, response-model baseline	Never cut, at any level of pressure
Uplift Random Forest, Optuna under a Qini objective, SHAP, drift monitors	Kept (SHOULD HAVE)
Azure deployment	Container deployment to Azure. A managed endpoint only if M5 is ahead
Neural estimator (shared-representation CATE)	Kept as the deep-learning component, time-boxed to one week, and the first item cut if M5 slips
Class-transformation trap figure	Kept: small, and it is Mohammed's demonstration
Causal forest, multi-treatment uplift, revenue uplift	Cut. Recorded as future work in the final report

One reassignment. The Streamlit dashboard moves from Mazen to Mohammed. With 3.5 weeks for implementation, Mazen carries the API, Docker, Azure, monitoring and CI/CD, and Mohammed's modelling work finishes in the second week of M5. Mohammed also draws the wireframes in M4, so the person who designs the screens builds them.

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4. Milestones

Milestone	Window	Exit condition
M1 Planning and management	to 16 Oct	Proposal, plan with Gantt chart, roles, risk register and KPIs committed; M1 retrospective written
M2 Literature review	16 Oct	Lecturer feedback received in the 16 October session and recorded, with the team's response to each point
M3 Requirements	13–16 Oct	Stakeholder analysis; user stories and use cases; functional requirements each traceable to a use case; non-functional requirements each with a stated way of measuring it
M4 Analysis and design	17 Oct – 6 Nov	Full DEPI design set; literature review with every source verified; split, harness and baselines built and tested; randomization report; central comparison reported with an interval
M5 Implementation	7 – 30 Nov	A clean checkout plus one documented command reproduces the headline result; the deployed endpoint answers a real request; the dashboard runs against it

Milestone	Window	Exit condition
M6 Testing and final	1 – 4 Dec	Champion confirmed on the locked test split, opened once; test cases and bug log; final report, user manual, presentation, infographic; rehearsed demo; final upload

5. Tasks

Every row is a GitHub issue in its milestone. The **owner** is the assignee and delivers the output. The **partner** reviews the pull request, or pairs, and can take the task over if the owner becomes unavailable. Task numbers continue from Milestone 1 (T1–T22).

M1 – M3 (due Friday 16 October)

The literature tasks T25–T29 were moved to M4 on 13 October (see section 1).

ID	Task	Owner	Partner	Due	Output
T22	Milestone 1 retrospective (#20)	Mahmoud	all	Fri 16 Oct	`docs/retrospective-m1.md`
T23	Project plan, Gantt chart and working agreement	Mahmoud	Ali	Thu 15 Oct	this file, `docs/working-agreement.md`
T24	Risk register: DEPI deadlines and team responsiveness	Mazen	Mohammed	Thu 15 Oct	`docs/risk-register.md`
T30	Lecturer review record (16 October session), and the requirements index page	Mahmoud	Aliaa	Fri 16 Oct	`docs/lecturer-review.md`, `docs/requirements/README.md`
T31	Stakeholder analysis	Mohammed	Mahmoud	Thu 15 Oct	`docs/requirements/01-stakeholders.md`
T32	User stories and use cases (use-case ID list posted Wed 14 Oct, 22:00)	Ali	Aliaa	Thu 15 Oct	`docs/requirements/02-user-stories-and-use-cases.md`
T33	Functional requirements	Mazen	Mohammed	Thu 15 Oct	`docs/requirements/03-functional-requirements.md`
T34	Non-functional requirements	Aliaa	Mazen	Thu 15 Oct	`docs/requirements/04-non-functional-requirements.md`
T35	Upload M1–M3 to the company repository	Mahmoud	:	Fri 16 Oct	company repository

M4 – System Analysis and Design (17 October – 6 November)

ID	Task	Owner	Partner	Due	Output
T36	Freeze the three interfaces: feature matrix and folds, metric signature, scoring response	Mahmoud	Aliaa, Ali, Mazen	Tue 20 Oct	`docs/design/interfaces.md` and typed stubs
T37	Data pipeline and joint stratified split, asserted and tested, both datasets	Aliaa	Mohammed	Tue 27 Oct	`src/netlift/data/`, tests
T38	Baselines on Hillstrom: random, blanket, response model	Mohammed	Ali	Fri 23 Oct	`src/netlift/models/`, MLflow runs
T39	Evaluation harness: Qini, AUUC, uplift@k, bootstrap intervals within arms, hand-computed fixture tests	Ali	Mahmoud	Fri 30 Oct	`src/netlift/evaluation/`, tests
T40	Randomization integrity report, both datasets (protected)	Aliaa	Ali	Fri 30 Oct	`reports/randomization-integrity.md`
T41	MLflow tracking server logging runs	Mahmoud	Mazen	Fri 30	`docs/experiment-`

ID	Task	Owner	Partner	Due	Output
	from the harness			Oct	`tracking.md` updated
T42	T-learner and X-learner, Hillstrom then Criteo	Mohammed	Mahmoud	Tue 3 Nov	`src/netlift/models/`, MLflow runs
T43	Permuted-treatment negative control	Ali	Mohammed	Wed 4 Nov	test and report section
T44	Uplift versus response model at equal budget, with interval: K1 (protected)	Ali	Mohammed	Fri 6 Nov	`reports/central-comparison.md`
#29	Sensitivity check: StratifiedGroupKFold on the feature-profile hash	Aliaa	Ali	Fri 6 Nov	report section
T45	API contract (OpenAPI) and FastAPI skeleton with mocked responses	Mazen	Ali	Fri 30 Oct	`src/netlift/api/`, `docs/design/api-contract.md`
T46	Use case diagram and descriptions	Ali	Mohammed	Fri 30 Oct	`docs/design/use-cases.md`
T47	Architecture, technology stack, component and deployment diagrams, deployment strategy	Mazen	Mahmoud	Wed 4 Nov	`docs/design/architecture.md`
T48	ERD, logical and physical schema	Aliaa	Mazen	Wed 4 Nov	`docs/design/data-model.md`
T49	DFD (context and level 1), wireframes for the four tabs, UI/UX guidelines	Mohammed	Mazen	Wed 4 Nov	`docs/design/dfd-and-ui.md`
T50	Problem statement and objectives; class, sequence, activity and state diagrams	Mahmoud	Aliaa	Wed 4 Nov	`docs/design/behaviour.md`
T51	Test plan: unit, integration and acceptance, keyed to requirement IDs	Mazen	Ali	Wed 4 Nov	`docs/design/test-plan.md`
T25	Literature: randomized experiments, non-compliance and the two datasets	Aliaa	Mazen	Wed 4 Nov	`docs/literature-review/01-experiments-and-data.md`
T26	Literature: meta-learners and class transformation	Mohammed	Aliaa	Wed 4 Nov	`docs/literature-review/02-meta-learners.md`
T27	Literature: uplift trees, causal forests and neural estimators	Mahmoud	Ali	Wed 4 Nov	`docs/literature-review/03-trees-and-neural.md`
T28	Literature: evaluating uplift models	Ali	Mohammed	Wed 4 Nov	`docs/literature-review/04-evaluation.md`
T29	Literature: machine-learning systems in production	Mazen	Mahmoud	Wed 4 Nov	`docs/literature-review/05-ml-systems.md`
T52	Align the specification's milestone and timeline sections with this plan	Mahmoud	Ali	Wed 4 Nov	specification `.tex` and PDF
T53	Assemble and submit the System Analysis and Design document; upload	Mahmoud	all	Fri 6 Nov	`docs/design/README.md`, company repository
T54	Milestone 4 retrospective and refinement of the M5 tasks	Mahmoud	all	Sat 7 Nov	`docs/retrospective-m4.md`

Diagram conventions, so nothing is drawn three times in three styles: Mermaid inside the Markdown file wherever Mermaid has the diagram type (class, sequence, state, ER, flowcharts for DFD and activity);

diagrams.net for use-case, component and deployment diagrams, committed as the `.drawio` source plus an exported `.svg`.

M5 – Implementation (7 – 30 November)

Refined at the M4 retrospective on 7 November. Feature freeze Friday 27 November; 28–30 November is fixes and documentation only.

ID	Task	Owner	Partner	Due
T55	Decision layer: policy modes, negative-uplift suppression, both abstention verdicts	Ali	Mazen	Fri 13 Nov
T56	Final meta-learners on Criteo and the class-transformation trap figure	Mohammed	Ali	Fri 13 Nov
T57	Uplift Random Forest and Optuna under a Qini objective; champion registered in MLflow	Mahmoud	Mohammed	Fri 20 Nov
T58	Production data pipeline (configuration, logging, errors) and the leakage audit enforced in CI	Aliaa	Mazen	Fri 20 Nov
T59	FastAPI service against the registered model, Docker image	Mazen	Ali	Fri 20 Nov
T60	PostgreSQL scoring log implementing the M4 schema	Aliaa	Mazen	Fri 20 Nov
T61	Campaign policy simulator with incremental-profit output, and SHAP	Ali	Mohammed	Fri 20 Nov
T62	Streamlit dashboard: four tabs against the API	Mohammed	Mazen	Fri 27 Nov
T63	Neural CATE estimator, time-boxed to one week, first to cut	Mahmoud	Ali	Fri 27 Nov
T64	Azure deployment, drift monitors, CI/CD complete	Mazen	Mahmoud	Fri 27 Nov
T65	Error analysis report and behavioural and invariance tests	Ali	Aliaa	Fri 27 Nov
T66	Data validation tests, dataset cards, data dictionary	Aliaa	Mohammed	Fri 27 Nov
T67	README, API documentation, one command that reproduces the headline result	Mazen	Aliaa	Mon 30 Nov
T68	Weekly end-to-end integration run on `main` (13, 20, 27 Nov); upload on 30 Nov	Mahmoud	Mazen	each Friday

M6 – Testing, Final Presentation and Reports (1 – 4 December)

Report drafts start in the last week of M5, so the final week is for assembly, not writing.

ID	Task	Owner	Partner	Due
T69	Final report: each member drafts their component section by Fri 27 Nov; assembly	Mahmoud	all	Thu 3 Dec
T70	Champion confirmed on the locked test split, opened once, procedure written down first	Ali	Mahmoud	Tue 1 Dec
T71	Test-case document and bug log	Mazen	all	Wed 2 Dec
T72	User manual and recorded demo video (fallback for the live demo)	Mohammed	Mazen	Wed 2 Dec
T73	Technical documentation: architecture, schema, API	Mazen	Aliaa	Thu 3 Dec
T74	Presentation, project infographic (DEPI template), per-member contribution slide	Mahmoud	Aliaa	Thu 3 Dec
T75	Two full demo rehearsals on the deployed system	Mahmoud	all	Wed 2 and Thu 3 Dec
T76	Final upload to the company repository	Mahmoud	:	Fri 4 Dec

6. Resource allocation

Member	GitHub	Area of ownership
Aliaa Gomaa	`aliaaelrefaey`	Data engineering and experiment validity
Mohammed Sherif	`mohamedshrief44`	Uplift modelling core, and the dashboard
Mahmoud Nadeem	`mahmoud-nadeem`	Advanced models and experiment tracking; team leader
Ali Alaa	`Ali-Alaa-Salah`	Evaluation, statistics and the decision layer
Mazen Mahmoud	`MazenSr`	MLOps, backend and deployment

(Earlier documents numbered these roles M1 to M5. The numbers are dropped here because M1 to M6 now name the milestones.)

The critical path is Aliaa then Ali: nothing is trustworthy until the split is valid, and no model can be compared with another until the harness with intervals exists. Everyone else builds against the frozen interfaces of T36 instead of waiting for them.

Commitment: 12 hours per member per week, about 90 hours each from 13 October to 4 December. Estimated load by milestone, in hours, to check the distribution is fair (estimates, revised at each retrospective):

Member	M1–M3	M4	M5	M6	Total
Aliaa	4	27	28	8	67
Mohammed	4	26	30	10	70
Mahmoud (incl. leadership)	7	29	30	14	80
Ali	6	28	34	10	78
Mazen	4	24	34	10	72

Partner reviews add roughly 10% to each figure. The margin between these estimates and the 90-hour commitment is the buffer for whatever the randomization report or the central comparison turns up.

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7. Tracking

What	Where
Tasks, owners, partners, due dates	GitHub Issues, grouped by milestone
Code and document review	GitHub Pull Requests
Individual contribution	Issue assignees, pull-request authors and reviewers, GitHub Insights
Decisions and the alternatives rejected	`docs/decisions/` (one numbered record per decision)
Daily progress, review requests, blockers	Team WhatsApp group
Weekly working blocks (12 hours per member)	Team WhatsApp group, posted at the Saturday meeting
Meetings and pairing	Discord voice channels
Repository activity feed	Discord `#github`
Experiment results	MLflow

Every task is an issue. Every issue closes through a pull request that names it. The chain from task to person to branch to review to merged commit is what makes each member's contribution provable at the final presentation, where DEPI asks each team to show who did what.

3. Task Assignment, Roles and Working Agreement

Task ownership is in section 2 (every task has an owner, a partner and a due date) and the resource allocation is in section 2.6. This section is how the team works day to day.

How the NetLift team works from 13 October 2026 to the final submission on 4 December 2026. It applies to every member, the team leader included. The plan it serves is `project-plan.md`.

Why this exists

Milestone 1 was delivered, but too much of it was spent waiting: for a review, for an answer, for someone else's piece to land. The daily routines we set up on Discord were not used, and there were no fixed times when the team was expected to be working or reachable. With seven weeks left and fixed DEPI deadlines, waiting is the one cost the schedule cannot absorb. Every rule below exists to remove a reason to wait.

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1. Where things happen

What	Where
Tasks, owners, partners, due dates	GitHub Issues
The work itself, and its review	GitHub Pull Requests
Decisions that others depend on	`docs/decisions/`, through a pull request
Daily standup, review requests, blockers, reminders	Team WhatsApp group
Meetings and pairing	Discord voice: `Team Meeting`, `Pair Room` (screen sharing)
Repository activity feed	Discord `#github`

The WhatsApp group is for short messages. Anything that has to be remembered goes into an issue or pull-request comment, where it stays attached to the work. Discord is kept for voice, because reviewing code together needs screen sharing, and for the `#github` channel, which shows repository activity as it happens. Its other text channels were removed.

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2. The weekly rhythm

Daily standup, by 22:00, Saturday to Thursday. Three lines in the WhatsApp group, with issue numbers:

```
Done:    T38 response model trained on Hillstrom (#NN)
Next:    T38 open the pull request
Blocked: nothing
```

It takes one minute. A day with no standup gets a check-in from the team leader the next morning.

Two meetings a week, 30 minutes each, in the Discord `Team Meeting` voice channel.

Meeting	When	Agenda
Plan the week	Saturday 21:00	The leader shares the GitHub status first: what is late, what is waiting for review. Each member says which issues will be in a pull request by Thursday.
Unblock	Tuesday 21:00	Anything that has waited more than 24 hours is resolved in the meeting, or reassigned.

The times are confirmed at the first meeting and then stay fixed. If you cannot attend, post your update in the group before the meeting starts.

Twelve hours per member per week, in blocks you announce. At the Saturday meeting each member posts their working blocks for the coming week in the WhatsApp group: at least three different days, no block shorter than 90 minutes, twelve hours in total. For example:

```
Sat 19:00-22:00 · Mon 20:00-23:00 · Wed 19:00-22:00 · Thu 16:00-19:00
```

The blocks tell the rest of the team when you can be reached for a review, a question or a pairing session, and they tell the leader when to check in. If a block moves, say so in the group. If a week will be shorter for you, say so on Saturday so the work can be moved before it is late, not after.

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3. Rules that remove waiting

1. **Build against the contract, not the person.** The three interfaces frozen in T36 (feature matrix and folds, metric signature, scoring response) exist so that nobody waits for anybody. If your input is not ready, work against a stub or fixture of the agreed shape and swap the real one in when it lands.
2. **Reviews within 24 hours.** When a pull request asks for your review, review it within 24 hours. If you cannot, say so in the group inside that window, and another member reviews it instead. Any member can approve; the partner is the first choice, not the only one. The reviewer checks each number against its source, and re-runs the script when the result depends on one. A pull request updated after review goes back to the reviewer the same day.
3. **Stuck for an hour, ask.** Post in the group: the issue number, what you tried, what you need. **Stuck for a day**, the team leader decides the same day: unblock it, simplify it, or move it.
4. **Make work visible early.** Open a draft pull request within 48 hours of starting a task, and push at the end of every working session. Work that is not pushed does not exist for anyone else.
5. **Small tasks.** Every task fits in three working days. If it does not, split it before starting.
6. **The internal deadline is the day before DEPI's.** The deadline day itself is for review, merging and uploading.

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4. When a task slips

Say it early. If you will miss a due date, say so in the group before the date passes, with a new date. That is normal, and the plan can absorb it when it is known in advance.

A silent slip is moved. If a task has no commit, pull request or message for 48 hours after its due date, the team leader moves it to its partner, and the original owner becomes the partner. This is not a penalty; it keeps the milestone alive. Both are recorded in the issue history.

DEPI's final evaluation asks each team to show what every member built and was responsible for. The GitHub record (issue assignees, pull-request authors, reviewers) is that evidence, for everyone.

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5. Definition of done

A task is done when all of these hold:

- Its pull request is merged into `main`, with CI green and approval from someone other than the author.
- The pull-request description starts with `Closes #N` for the issue it resolves.
- Documentation is updated wherever the change alters behaviour, a number or an interface.
- Every number in a document comes from a script or file in the repository, not from memory.

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6. Decisions

A decision other people depend on (a library, an interface, a threshold, a scope cut) gets a numbered record in `docs/decisions/`, in the format of `0001-uplift-library.md`: context, options considered, decision, consequences. An agreement in the WhatsApp group is not a decision until it is recorded there.

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7. The team leader

The team leader runs both meetings, shares the GitHub status every Saturday, decides on anything blocked for a day within that day, reviews when nobody else can, and keeps the plan current. The leader is held to every rule above.

4. Risk Assessment and Mitigation Plan

Rating scale

Likelihood

- **Low**, unlikely, but has happened on projects like this
- **Medium**, more likely than not to happen at least once
- **High**, expect it to happen

Impact

- **Low**, costs hours
- **Medium**, costs days, or weakens one deliverable
- **High**, threatens a milestone or a headline claim

Reviewed at every milestone boundary. Owners update their own rows.

ID	Risk	Category	Likelihood	Impact	Early warning signal	Mitigation	Owner	Status	Last reviewed
R01	The central hypothesis does not hold: uplift targeting does not outperform the response model on the project data.	modelling	Medium	High	The uplift-vs-response comparison due Fri 6 Nov shows no meaningful advantage over the response-model baseline.	Run the comparison by Fri 6 Nov and investigate the result early rather than waiting until the final report.	Mohammed	open	2026-10-14
R02	Effects are too small to distinguish from statistical noise.	modelling	Medium	High	Bootstrap confidence intervals for the headline uplift result include zero or remain too wide to support a clear conclusion.	Evaluate both visit and conversion outcomes, report confidence intervals, and use the abstention path when evidence is inconclusive.	Ali	open	2026-09-29
R03	Curriculum gap because the project does not include computer vision or NLP.	schedule	Medium	High	Instructor feedback indicates that CV/NLP coverage is expected or may affect evaluation.	Raised explicitly during the Week 7 review; the project was approved as it stands. No further mitigation is required unless the scope changes.	Mahmoud	resolved	2026-09-29
R04	Uneven team workload leaves some members without substantial ownership.	team	Medium	High	A member has no clearly owned component, deliverable, or contribution to the final demo.	Maintain independent workstreams and require every member to own a component that appears in the final	Mahmoud	open	2026-09-29

ID	Risk	Category	Likelihood	Impact	Early warning signal	Mitigation	Owner	Status	Last reviewed
						system/demo.			
R05	Metric overfitting caused by repeatedly tuning against the locked test split.	modelling	Medium	High	Test-set results are used to make additional modelling or hyperparameter decisions after the split has been opened.	Lock the test split, document the rule before opening it, and use validation/development data for tuning.	Ali	open	2026-09-29
R06	Silent data leakage contaminates uplift estimates.	data	Medium	High	A forbidden feature/column appears in the modelling data or the permuted-treatment negative control produces an unexpected effect.	Automate forbidden-column checks and run the permuted-treatment negative control.	Aliaa	open	2026-09-29
R07	Cross-fitting is implemented incorrectly in the X-learner or R-learner.	modelling	Medium	High	A test shows that training and prediction folds overlap or fold membership does not match the intended cross-fitting design.	Prefer a tested library implementation and add tests that explicitly verify fold membership.	Mohammed	open	2026-09-29
R08	Scaling problems when processing the 14M-row Criteo dataset.	engineering	Low	Medium	Full-data processing exceeds available memory, becomes impractically slow, or fails during loading/preprocessing.	Use Parquet with selected columns, chunked preprocessing, and a stratified subsample for iteration; use the full dataset for final reporting.	Aliaa	open	2026-09-29
R09	Azure/cloud credits are exhausted before deployment and experimentation are complete.	engineering	Low	Medium	Cloud spending approaches the configured budget threshold or compute resources remain running after a session.	Train locally where practical, shut down compute after each session, and configure a budget alert below the credit limit.	Mazen	mitigating	2026-09-29
R10	Live demo infrastructure fails on presentation day.	engineering	Low	Medium	The deployed endpoint or dashboard cannot be reached or fails during a rehearsal.	Maintain a recorded demo and a local fallback mode that can run without the cloud endpoint.	Mazen	open	2026-09-29
R11	The development environment works on one machine but fails on another team member's machine or in CI.	engineering	Medium	High	`verify_environment.py` produces different results across team members, or a dependency/version mismatch appears between machines or CI.	Pin the environment, run the verification script on every team member's machine, and require all five members to produce passing results.	Mazen	mitigating	2026-09-29
R12	Environment verification gives a false passing result because a required dependency is pinned but not actually checked.	engineering	Medium	High	A required package is present in the environment specification but missing from `verify_environment.py` checks, allowing the script to report success despite incomplete verification.	Keep the verification checks synchronized with the pinned environment and update the check count whenever a required dependency is added.	Mazen	mitigating	2026-09-29
R13	Branching, pull-request review, or merge workflow	engineering	Medium	Medium	A task branch is not correctly pushed/selected for	Follow the agreed branch/PR workflow, verify the correct	Mazen	mitigating	2026-09-29

ID	Risk	Category	Likelihood	Impact	Early warning signal	Mitigation	Owner	Status	Last reviewed
	delays integration of completed work.				comparison, a required review remains unresolved, or a PR cannot be merged when the task is otherwise complete.	base and compare branches, resolve review conversations promptly, and require approval before merging.			
R14	A team member becomes unavailable for an extended period.	team	Medium	High	Two missed standups in the team WhatsApp group, or a task is 48 hours past due with no commit, PR or message.	Every task has a named partner who can take it over. If the owner is unavailable, the team leader transfers the task to the partner after 48 hours and records the change in the issue history; raise the issue in the team channel within two days rather than absorbing it silently.	Mahmoud	open	2026-10-14
R15	The company repository is never granted, leaving the project on a personal account.	engineering	Medium	Low	The company has not responded about repository access.	Transfer the repository to the organisation if required; preserve the Discord `#github` channel as the parallel record either way.	Mahmoud	open	2026-10-14
R16	A critical project milestone slips, reducing the time available to react to its findings.	schedule	Medium	High	The randomization-integrity report is not on track for Fri 30 Oct, or the uplift-vs-response comparison is not on track for Fri 6 Nov.	Track the two protected dates explicitly, review progress at each milestone boundary, and protect Fri 30 Oct and Fri 6 Nov before lower-priority work.	Mahmoud	mitigating	2026-10-13
R17	DEPI deadlines remove approximately six weeks from the original project schedule, leaving insufficient time for the full planned scope.	schedule	High	High	Any M5 task is past its due date in `docs/project-plan.md`, or the 27 Nov feature freeze is at risk.	Follow the scope order in `docs/project-plan.md` section 3; write tests alongside implementation during M5 rather than deferring testing; enforce the feature freeze on 27 Nov.	Mahmoud	mitigating	2026-10-14

5. Key Performance Indicators (KPIs)

Two kinds of KPI are used and kept distinct. **Pass/fail** is a condition that is either met or not. **Measured** means a number reported with a 95% bootstrap confidence interval and no numeric target set in advance, because there is no prior result on this data to set one from. Where a success condition exists for a measured KPI, it is structural, such as an interval that excludes zero.

KPI	How it is measured	Type	Success condition	First reported	Owner
K1: Incremental conversions vs. response model (primary)	Simulated campaign replay (Spec Section 14) on the locked test split: both policies select the same number of customers under the same budget; incremental conversions are measured against the withheld slice.	Measured (95% interval, no preset target)	The 95% bootstrap interval of the difference (uplift policy minus response model) excludes zero in favour of the uplift policy. An interval containing zero is a reportable finding, not a failed project.	M4 (Week 12-13)	Ali
K2: Qini coefficient and AUUC	Evaluation harness, on the locked test split, with 1000+ bootstrap resamples within treatment arms. Library and settings must be named in the report (Qini/AUUC normalisation differs across libraries).	Measured (95% interval, no preset target)	The 95% bootstrap confidence interval is entirely above zero, indicating performance demonstrably better than random targeting. (An interval entirely below zero would indicate worse-than-random targeting, not success: “excludes zero” alone is not sufficient.)	M4 (Week 12-13)	Ali
K3: Incremental profit per unit of spend	Replay output (K1) converted using the stated per-contact cost and margin (Spec Section 13).	Measured (95% interval, no preset target)	Reported with a confidence interval, and with the cost and margin assumptions stated on the same line as the figure.	M4 (Week 12-13)	Ali
K4: Negative control	Full pipeline re-run on a randomly permuted treatment indicator.	Pass/fail	The 95% interval of the measured uplift contains zero.	M4, re-run at M6	Ali
K5: Calibration	Predicted vs. measured incremental conversions per decile, in the replay.	Measured (95% interval, no preset target)	Differences straddle zero across deciles rather than showing a systematic direction (over- or under-prediction).	M5	Ali
K6: Randomization integrity	Standardized mean difference (SMD) per covariate, plus the ROC-AUC of a classifier trying to predict the treatment flag from the covariates alone.	Pass/fail	Every /SMD/ < 0.1 (the conventional threshold, fixed in config before the test is run) and treatment-predictability AUC near 0.5. Any breach is documented rather than smoothed over.	M3 (Week 9-10)	Aliaa
K7: Leakage control	Automated forbidden-column check in CI, plus the K4 negative control as a second, independent check.	Pass/fail	The check fails the build when a post-treatment column reaches the model matrix, and a unit test asserts that it does.	M4	Aliaa
K8: Reproducibility	Clean checkout, one documented command, output compared against the recorded headline number.	Pass/fail	Reproduces exactly, from config plus the git commit hash.	M5	Mahmoud

KPI	How it is measured	Type	Success condition	First reported	Owner
K9: Serving performance	Load test against the deployed endpoint.	Pass/fail	Team-set target, to be revised once first measured: p95 under 300 ms for a single score; a one-million-row batch inside ten minutes on the project's own hardware. (Target as stated in Spec Section 46: not invented here.)	M5	Mazen
K10: Engineering quality	CI on every pull request: lint, type check, unit, data, model-behaviour, integration and API tests.	Pass/fail	Green on the merge commit of every pull request.	M5-M6	Mazen
K11: Documentation completeness	Checklist of the DEPI-required artefacts (Spec Section 45).	Pass/fail	Every required artefact exists and is current.	M7	Team leader